

1. **NAME:** -**Prof. Dr. GHORUDE TATYARAO NAMDEVRAO**

2. **DESIGNATION:** (**Professor**)

Vice-Principal & Head, Department of PHYSICS

GOKHALE EDUCATION SOCIETY'S

N. B. Mehta Science College, Bordi

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3. **EDUCATIONAL QUALIFICATIONS :**

Degree	Board/University	Year
Ph. D. (Electronic Science)	University Of Pune, Pune	2008
M.Sc.	Gulbarga University, Gulbarga	1991
B.Sc.	Gulbarga University, Gulbarga	1988

4. **ORIENTATION COURSES :**

1. UGC Sponsored Orientation course 2nd Dec. to 30th Dec.2014 – BAMU, Aurangabad.
2. Orientation course 2nd Feb to 11th Feb 1999 – Tata Institute of Social Science, Chembur, Mumbai
3. Teacher Orientation Programme on 29th Jan. 2006 - Dr. MSG Foundation Mumbai in Collaboration with G.E. Society.

5. **REFRESHERS COURSES :**

1. Refresher course Organized by ASC Mumbai University during 14th Feb. to 10th Mar. 2000.
2. Refresher course Organized by ASC Mumbai University during 01st Dec. to 28th Dec. 2000.
3. Refresher course Organized by ASC Pune University during 03rd March to 23rd Mar. 2005.
4. Refresher course on “**RESEARCH METHODOLOGY and DATA ANALYSIS**”
(Sharpening skills through MSEXcel, SPSS, MATLAB and R) Organized by Ramanujan College, University of Delhi under the aegis of Ministry of Education during 24th March. 2021 - 07th Apr.2021.
5. Refresher course on “**Managing Online Classes and Co-creating MOOCs 25.0**”
Organized by UGC-HRDC, conducted by Ramanujhan college, Delhi from 07th April to 21st April 2023.

6. SHORT TERM COURSE :

- 1) Short term course on “**Role of a Teacher in Enhancing Student Potential**” Organized by UGC Human Resources Development Centre, University Of Mumbai on 5th Oct. 2015-10th Oct.2015.
- 2) Short term course on “**SAFETY AND HYGIENE IN (CYBER) WORLD**” Organized by Ramanujan College, University of Delhi under the aegis of Ministry of Education during 01st March. 2021 - 08th March.2021.
- 3) Short term course on “**Academic Administration**” Organized by UGC - HRDC, University Of Mumbai on 9th Mar. 2023-15th Mar.2023.
- 4) Short term course on **19-NEP 2020 Orientation Sensitization Programme (DEC-A) under Malviya Mission Teacher Training Programme (MM-TTP)** of UGC organized by BAMU, Chatrapati Sambhaji Nagar from **9 th to 18 th Dec 2024**.

7. WORKSHOPS & SEMINARS ATTENDED :

1. Worked as **resource person** for the Physics teacher’s workshop of **Orienting towards NEP 2020 compliant syllabus & curriculum** on 28/10/2024 at Sathaye College, Vile Parle.
2. Participated in 5days (29th May to 2nd June 2023) E-workshop on “Scientific communication for researchers” Conducted by Research and Development centre, BMS engineering college, BANGALORE
3. Completed Covid – 19 awareness programs organized by NSS & Disaster management centre Shivaji university Kolhapur on 30/04/2020 with 80% score.
4. Attended webinar on “Strengthening college education” by M. V. Mandali’s college on 09 May 2020
5. Participated in the webinar on “Supercapacitors, Solar cell devices & solar thermal technology” during 11-14 May 2020 Organized by Shri Ramdeo Baba College of engineering and management, Nagpur.
6. Participated in the international webinar series on ‘Atomic force microscopy and it’s applications’ on 13 June 2020, organized by R. D. National college Mumbai.
7. Participated in the international webinar on ‘Career at DRDO’ by R. D. National college, on 16 June 2020.
8. Attended international webinar on ‘Sixty years of LAESR, organized by Jamshedpur Women’s college on 26th June 2020.
9. Participated RUSA sponsored national webinar conducted by Patkar college, Mumbai on 10th July 2020
10. Participated in one day national level webinar on ‘CAMERA – Physics and Electronics’ organized by Royal College Mumbai on 20 July 2020.

11. Participated in the international symposium on advanced research in Physics during 27-31 July 2020 organized by Department of Physics, Bangalore University.
12. Participated in the one-day national webinar on 'Impact of pandemics on Education' organized by Sathaye college Mumbai on 31st July 2020.
13. Attended the RUSA – Sponsored state level webinar UGC'S career advancement scheme (CAS) organized by Patkar college Mumbai on 8th January 2021.
14. Participated in the national conference on 'Sustainable development – A green approach' organized by Thakur college Kandivali, & University of Mumbai, supported by ABARD on 6 March 2021.
15. Participated in national webinar on 'Execution of national education policy 2020 in pandemic panorama' organized by K. G. Mittal College, Malad, Mumbai on 24th July 2021.
16. Participated in the one-day national workshop on 'XRD Instrumentation, Analysis, Interpretation of spectra and applications' organized by Birla college on 28th Aug 2021.
17. Attended one day workshop on 'Research Tools' organized by G.E.Society's HPT arts & RYK Science college Nasik on 29th September 2021.
18. Participated in Two days national webinar on ' Career Advancement Scheme and writing skills of review of literature and framing the objectives and hypotheses' organized by G. E. Society's Arts, commerce and Science college, Shreewardhan on 7th and 8th Oct 2021
19. Participated in a webinar National Education policy 2020 – it's implementation in Higher education organized by Nagin das khandwala college Mumbai on 9th Oct 2021.
20. Participated in one day state level webinar on 'New Education policy 2020' organized by Gholap and Pawar college Sivle, Murbad. On 16th Oct 2021.
21. Attended one day National webinar on Journey towards Nuclear Energy Organised by ASP College Devrukh, Ratnagari on 27th Feb 2022.
22. Workshop on S.Y.B.Sc. organized by Dept. of Physics, St.- Xaviers college, Mumbai On 26th Nov. 1997
23. Workshop on T.Y.B.Sc. Organized by Dept. of Physics, D.G. Ruparel college, Mahim Mumbai On 08th and 09th Aug. 1998.
24. Workshop on T.Y.B.Sc. Organized by Dept. of Physics, Mumbai University On 11th and 12th Feb. 1999.
25. Workshop on S.Y.B.Sc. Organized by Dept. of Physics, D.G. Ruparel college, Mahim Mumbai On 08th Feb. 2003.

26. Workshop on Device grade material development using ion Beam Organized by Nuclear Science Centre, New Delhi and Dept. of Physics University of Mumbai On 03rd and 04th Nov. 2003
27. Workshop on S.Y.B.Sc. Organized by Dept. of Physics, Bhavans College, Andheri On 10th Aug. 2004.
28. State Level Workshop on “RTI” act 2005 in Higher Education Organized by Public Night Degree College Vakola, Santacruz On 12th April 2008.
29. Workshop on Project management Using MS project tool Organized by SRG consultant and Dept. of Electronic Science University of Pune On 04th and 05th May 2008.
30. Workshop on IQAC and Reaccreditation Organized by QAC-NAAC, Govt. of Maharashtra and S.D.S.M. College Palghar On 30th Aug. 2008.
31. Seminar on Examination System and Procedures Organized by N.B.Mehta Science College, Bordi On 24th Feb. 2007.
32. Worked as **chairperson** in international multidisciplinary conference at G.S.M College in association with I NAAR on 26th March 2016 in the subject of science.
33. **Editorial Board Member** of Research Journal of Physical Sciences, International peer reviewed monthly journal **Aug 2016 onwards.**
34. Worked as **V.C. subject expert nominee** for CAS Interview in Siddhartha College, Mumbai on 16th Sept. 2016.
35. **Dr.T.N.Ghorude worked as chairperson in science session at conference in Vadkun College at dahanu on 9th March 2019.**
36. **Dr.T.N.Ghorude worked on advisory committee and as chairperson for four sessions in International Physics Conference (I – CAN_N 2019-20) at S.D.S.M’s college Palghar during 26 Nov to 28 Nov 2019**
37. Worked as **Convener** in an online awareness seminar ‘**Path to Scientific Research**’ organised by **We Mentor Research (WeMR)** at N. B. Mehta (Valwada) Science College, BORDI on 28th February 2024.
38. Worked as **Chairperson** for National Conference on Management and Technology (NCMAT) on 15th June 2024 organised by Deccan Education Societys Navinchandra Mehta Institute of Technology and Development, DADAR
39. Worked as Convener in an online skill Development workshop on ‘**Writing skills for resume Building**’ organised by **We Mentor Research(WeMR)** at N. B. Mehta(Valwada) Science College, BORDI on 30th Aug 2024.

8. EXPERIENCE: 30 years (UG) and 9 years (PG) teaching experience and 24 years Research experience.

9. AREA OF SPECIALIZATION : Solid State Physics, Optoelectronics, Sensors and Transducers, Polymers, etc

10. Research Publications in International Journals:

1. Vinod S. More, M. P. Patil, R. P. Tandel, **T.N. Ghorude**, Gitesh G. Padhye published research paper entitled, **“Study of Structural, Morphological, Electrical and Thermal Characteristics of PPy-Fe(ClO₄)₂-Au Nanocomposite and its Application in Ammonia Gas Sensing”** in *Indian Journal of Science and Technology(IJST)* (**UGC Care listed Journal**). Vol-17, Issue-46, 18th December 2024, Page No. 4842-4853.
2. Raju P. Tandel, Vinod S. More, **T.N. Ghorude** has published paper entitled **“INVESTIGATING THE ROLE OF FE ON STRUCTURAL, MORPHOLOGICAL AND ELECTRICAL CHARACTERISTICS OF POLYPYRROLE NANOPARTICLES”** in *International Journal of Scientific Research*. ISSN: 2277-8179, Volume-13, Issue-08, August-2024, pp 01-03, Impact Factor 5.711
3. Vinod S. More, **T.N. Ghorude**, Raju Tandel, B.K. Sakhare, G.G. Padhye has presented paper entitled **“From Iron to Sensing: Investing the Role of Iron Chloride in Polypyrrole based LPG Gas detection”** in one day National conference on **“Recent Developments In Basic And Applied Sciences (RDBAS 2024)”** organized by Dr.S.D.D. Arts, commerce and Science college, Wada, Palghar on Saturday, 2nd March 2024. Also, Published in **Research Journey**, International E- Research Journal, E-ISSN 2348-7143, Impact Factor – 6.625, **Special Issue 338 : Recent Developments in Basic and Applied Sciences**, March 2024, page No. 194-202.
4. Vinod S. More, **T.N. Ghorude**, Apurva M. Patil has presented paper entitled **“Synthesis of Polypyrrole Nanoparticles for enhanced H₂S Gas Sensor Fabrication”** in one day **International Multidisciplinary e-conference on Recent Trends in Science, Commerce & Humanities (IMCRTSCH-2024)”** organized by D.V. College, Goregaon, Raigad on Saturday, 16th March 2024. Published in **International Multidisciplinary E-Journal**. ISSN 2277-4262, **Impact Factor – 8.285**, Vol-13, Issue-05, 01st May 2024, page No. 237-246.
5. Minal P. Patil, Gaurav Singh, Vinod S. More, **T.N. Ghorude**, **“DEVELOPMENT OF MICROCONTROLLER BASED HAEMOGLOBIN METER”** published in

International Journal of Scientific Research. ISSN: 2277-8179, Volume-13, Issue-02, February 2024, pp 47-49, Impact Factor 5.711

6. B. K. Sakhare, **T.N. Ghorude**, V.S. More, **“Polypyrrole Infused Cotton Fabrics: An Elegant Solution for Gas Sensing”** published in **International Journal of Scientific Research (IJSR)** ISSN: 2277-8179 Volume-12 Issue-11, November 2023 page No.66-68, Impact factor 5.711
7. Sahebrao B. Pagar , **Tatyrao N. Ghorude** , Maria P. Nikolova , K. SenthilKannan **“Synthesis, physical, chemical, biological, mechanical and electronic studies of polypyrrole (PPy) of versatile scales for electro-mechano, pharmaceutical utilities”** Science Direct **Heliyon** Volume-9, Issue-9, <https://doi.org/10.1016/j.heliyon.2023.e20086> Received 17 May 2023; Received in revised form 24 August 2023; Accepted 11 September 2023 Available online 12 September 2023 Impact factor 3.776
8. Sahebrao. B. Pagar, Dr. Minal. P. Patil, **Dr. Tatyrao. N. Ghorude** **“quantative color measurement of ph indicator papers using developed tristimulus colorimeter and hr-4000 high resolution spectrophotometer”** [International journal of scientific research](#) Volume - 12 | Issue - 09 September - 2023 | PRINT ISSN No. 2277 - 8179 | DOI : 10.36106/ijsr Impact Factor 5.711
9. Sahebrao B. Pagar, **Tatyrao N. Ghorude**, Mrinalini D. Deshpande, K. SenthilKannan **“Facile Fabrication of Room Temperature Based H₂S Gas Sensor Using ZTO-Ag@PPy Hybrid Nanocomposite”** **Springer** Journal of Inorganic and Organometallic Polymers and Materials <https://doi.org/10.1007/s10904-023-02650-8> published online: 24 May 2023 Impact factor 3.518
10. Vinod S. More , **T. N. Ghorude** , Supriya N. Save , B. K. Sakhare , R. P. Tandel **“Quantification of Fe (II) Ions in the Synthesis of Polypyrrole by Spectrophotometric Detection”** International Journal of Science and Research (IJSR) ISSN: 2319-7064 Volume 12 Issue 4, April 2023 page No. 1812 – 1816 Impact factor 7.942
11. **Dr. T.N.Ghorude** ,N. S. Shah, V.D.Thakur **“MICROCONTROLLER BASED FLAT-FOOT DETECTOR USING LOAD CELLS”** INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH , Volume - 11 | Issue - 09 | September - 2022 | PRINT ISSN No. 2277 - 8179 | DOI : 10.36106/ijsr Page No. 13-14. Impact Factor 5.862
12. R. Mundotia , **T. Ghorude** , D.C. Kothari c , A. Kale b , N. Thorat **“Tuning bilayer period of AlN/CrN super lattice coatings deposited using cathodic arc technique for superior mechanical properties and thermal stability”** ScienceDirect **ELSEVIER** Applied Surface Science Advances Volume 7, February 2022, Impact Factor-6.2

13. **Dr. T.N.Ghorude**, G. G. Singh “ **μ C based Fuel quantity and density measurement**” International Journal of All Research Education and Scientific Methods (IJARESM), ISSN: 2455-6211 Volume 9, Issue 10, PP 351-354 October -2021, Impact Factor: 7.429,
14. **T. N.Ghorude** and M. P. Patil “**SPECTRAL RESPONSES OF C.I.E.1964 R.G.B. DATA USING PYTHON**” International Journal of Advance and Innovative Research ISSN: 2278-7844, Volume 8, Issue 2 (III), April - June 2021. Impact Factor-7.15
15. R. Mundotia, **T. Ghorude**, A. Kale and U. Mhatre. “**Study of multilayer and multi-component coatings deposited using cathodic Arc technique on H-13 hot work steel for die-casting applications**” **International Journal of Nano Dimension**, Volume 11, Issue 2, April 2020, Pages 177-187
16. R. Mundotia, D.C. Kothari, A. Kale, U. Mhatre, K. Date, N. Thorat, **T. Ghorude**. ‘ **Effect of ion bombardment and micro-blasting on the wear resistance properties of hard TiN coatings**’ **ELSEVIER Materials Today: Proceedings** Volume 26, Part 2, 2020, Pages 603-612. Impact Factor-31.041
17. R. Mundotia, N. Thorat, A. Kale, U. Mhatre, D. C. Kothari, T. Kovacs and **T. Ghorude**. “**Study of corrosion properties of CrN and multilayer CrN/Cr coating at different electrolyte temperatures deposited on stainless steel by vacuum arc process**” **AIP Conference Proceedings** Volume-2115, Issue-I, July-2019.
18. M.P.Patil and **T.N.Ghorude** “**pH Indicator paper Color measurement using wireless tristimulus colorimeter**” International Journal of Scientific Research Volume-7, Issue-7, July-2018, ISSN 2277 – 8179, IF: 4.758, IC Value: 93.98 Pages 525-526.
19. M. P. Patil and **T. N.Ghorude** “**Measurement of hemoglobin (hb) in blood using photo detector by reflection method**” International Journal of Scientific Research Volume-6, Issue-9, September-2017, ISSN No 2277 - 8179, IF : 4.176 | IC Value : 78.46 Page No. 583-584.
20. **T.N.Ghorude** and M.P.Patil “**CIE standard 23 shades Color Measurement Using Wireless Tristimulus Colorimeter**” **International Journal of Chemical and Physical Sciences**, ISSN:2319-6602 IJCPS Vol. 5 Special Issue Nov. 2016 pp 15-20.
21. **M.P.Patil** and **T.N.Ghorude** “**Identification of Biological colonies using wireless digital colorimeter**” International Journal of Advances in Management Technology and Engineering Sciences Volume-6, Issue 2(1), March 2016 PP 18 – 21.
22. Mrs. Chinmaya Bari and **Dr. T.N.Ghorude** “**FACE RECOGNITION USING SKIN COLOR SEGMENTATION OF YCBCR AND RGB COLOR MODELS**” International Journal of Emerging Technology in Computer Science & Electronics (IJETCSE) ISSN: 0976-1353 Volume 23 Issue 1 –JUNE 2016. Impact Factor-4.78

23. **T. N. Ghorude**, D. B. Gaikwad and A. D. Shaligram “**Microcontroller Based Color Measurement Using RGB LEDs**” International Journal of Scientific Research ISSN-2277-8179 Vol. 1 issue 3 Aug 2012, Page No. 93-95. Impact Factor-5.711
24. **T.N.Ghorude**¹, Apurva M. Patil², Urmila L. Chimada³ “**Development of Hazardous gas detection system using Arduino R3**” International Journal of All Research Education and Scientific Methods (IJARESM), ISSN: 2455-6211, Volume 9, Issue 10, October -2021, Impact Factor: 7.429

International conferences

25. Vinod S. More¹, **T. N. Ghorude** has presented paper entitled “**From Particles to Fibers: Harnessing Polypyrrole's Conductivity for Gas Sensing Applications**” in one day International Conference on **Recent Advances in Material Sciences (RAMS-2024)** organized by ASP college, Devrukh
26. Vinod S. More, **T.N. Ghorude** “**Study of Structural and electrical properties of surface modified PPy-AuNPs thin film for gas sensing application**” presented in International Conference on ‘Advanced Materials for Physical, Chemical and Biological Applications’ funded by ‘Rashtriya Uchchatar Shiksha Abhiyan,’ jointly organised by the K.B.P. College, Vashi on 03rd & 04th March 2023.
27. Bhimrao Sakhare, R.P. Tandel, **T.N. Ghorude**, G.G. Padhye “**Synthesis and Characterization of Polypyrrole composites using various dopants**” presented in International Conference on ‘Advanced Materials for Physical, Chemical and Biological Applications’ funded by ‘Rashtriya Uchchatar Shiksha Abhiyan,’ jointly organised by the K.B.P. College, Vashi on 03rd & 04th March 2023.
28. Vinod S. More, Bhimrao Sakhare, Raju Tandel, Gitesh G. Padhye, **T. N. Ghorude** “**Smart Conducting Fabrics: Applications in Bio-Medical Sciences**” presented in International Conference on Recent Innovations in Science and Technology-2023 (RIST-2023) at I.Y. college between 10th and 11th Jan 2023
29. Bhimrao Sakhare, Raju Tandel, Vinod S. More, Gitesh G. Padhye, **T. N. Ghorude**, “**Synthesis of Polypyrrole based Smart Conducting Fabrics: Applications in Bio-Medical Sciences**” presented in International Conference on Recent Innovations in Science and Technology-2023 (RIST-2023) at I.Y. college between 10th and 11th Jan 2023
30. Raju Tandel, Bhimrao Sakhare, Vinod S. More, Gitesh G. Padhye, **T. N. Ghorude**, “**Electrical Properties of Polypyrrole Conducting Polymer at Various Dopant Concentrations**” presented in International Conference on Recent Innovations in Science and Technology-2023 (RIST-2023) at I.Y. college between 10th and 11th Jan 2023

31. **T.N.Ghorude¹**, Apurva M. Patil², Urmila L. Chimada³ **“Development of Hazardous gas detection system using Arduino R3”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology jointly organised by Sonopant Dandekar College Palghar and Dept of Physics University of Mumbai between 26th to 28th Nov 2019.
32. Rajesh Mundotia^{1,2}, Ashwin Kale², Umesh. Mhatre², **T. N. Ghorude¹** **“Study of Multilayer and Multi-Component Coatings Deposited Using Cathodic Arc Technique on H-13 hot work steel for Die- Casting Applications”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology at Sonopant Dandekar College between 26th to 28th Nov 2019.
33. **T. N.Ghorude¹** and M. P. Patil² **“SPECTRAL RESPONSES OF C.I.E.1964 R.G.B. DATA USING PYTHON”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology jointly organised by Sonopant Dandekar College Palghar and Dept of Physics University of Mumbai between 26th to 28th Nov 2019.
34. **T.N.Ghorude¹** G. G. Singh² Kapil Wag^{h3} **“ μ C based Fuel quantity and density measurement”** Presented in International Conference on Advanced Nanomaterials & Nanotechnology jointly organised by Sonopant Dandekar College Palghar and Dept of Physics University of Mumbai between 26th to 28th Nov 2019.
35. RAJESH MUNDOTIA ,Others & **Dr.T.N.Ghorude.** **“Effect of ion bombardment and micro-blasting on the surface profile of TiN coated HSS samples deposited by Cathodic Arc Physical Vapour deposition (CA-PVD) technique”** in International Conference on Thin Film, November 13 to 17. 2017. at CSIR
36. **M.P.Patil** and T.N.Ghorude **“Identification of Biological colonies using wireless digital colorimeter”** international multidisciplinary conference at G.S.P College Talasari in association with **INAAR** on 26th March 2016.
37. **T. N.Ghorude** and M.P.Patil **“CIE standard 23 shades Color Measurement Using Wireless Tristimulus Colorimeter”** International Conference On Photon: Multiple & Creative Solution to Challenges (ICPMCSC) Organized by Dept. Of Physics Ratnam college, Mumbai 4th and 5th Dec.2015.
38. **T. N. Ghorude** and A. D. Shaligram **“Wireless online Color measurement using TCS 230 Color Sensor”** 3rd International Science Congress (ISC-2013) Conference at **Karunya University, Coimbatore**, Tamil Nadu, India from 8th - 9th December 2013.
39. **T. N. Ghorude**, A. L. Chaudhari and A. D. Shaligram **“Quantitative Color Measurement of pH Indicator paper Using Trichromatic LEDs and TCS230 Color Sensor”** International conference on Experimental Mechanics (ICEM08) Organized by The Chinese

Society of Theoretical and Applied Mechanics (CSTAM), **Southeast University, Nanjing, China (SEU) during 08th – 11th Nov. 2008 at Nanjing, CHINA - Presented Orally**

40. **T. N. Ghorude** and A. D. Shaligram “**Skin Color measurement using Tristimulus LEDs and TCS 230 Color sensor**” International conference on Microwaves and Optoelectronics (ICMO2007) Organized by Dr. B.A.M. University, Aurangabad, December 17-20, 2007.

Research Publications in National Journals/conferences

41. Vinod S. More, **T. N. Ghorude** has presented paper entitled “**Development of Low ppm Ammonia Gas Sensor using Polypyrrole-Gold Nanoparticles**” in one day **National Conference on “Sustainable Development in Society, Economy and Science”** organized by St. John College, Palghar on Saturday, 4th May 2024.
42. Vinod S. More, B.K. Sakhare, R.P. Tandel, G.G. Padhye, **T.N. Ghorude**, “**Electric Elegance: Enhancing Cotton Fabric with Conducting Polypyrrole for Gas Sensing Applications**” **accepted for publication in Mapana Journal of Sciences**, comes under **UGC Care listed journals**
43. **B.K. Sakhare**, V.S. More, T.N. Ghorude, “**Spectrophotometric Approach to Quantify the Iron(II) Ions in the Synthesis of Polypyrrole**” published in **Cosmos Multidisciplinary Research E-Journal**. ISSN: 2456-1665 Volume 8 Issue 10, October 2023 page No. 09-15, Impact Factor 5.3.
44. Vinod. S. More, **T.N. Ghorude**, M.P. Patil, B.K. Sakhare, R.P. Tandel, G.G. Padhye, “**Comprehensive Analysis of γ -radiation Impact on PVA/PTH Composite Films for Enhanced Battery Functionality**” published in **Indian Journal of Science and Technology**. ISSN: 0974-6846 Volume 16 Issue 46, 13 December 2023, page No. 4349-4357.
45. **T. N. Ghorude** and A. D. Shaligram “**Microcontroller driven RGB LED system for tristimulus surface colorimetry**” Indian journal of Applied Research, ISSN: 2249-555X, Volume-1, Issue-5, February 2012, Page No.81-83.
46. Chinmaya Bari and **T. N. Ghorude** “**Artificial Neural Network Based Pattern Recognition System**” National Conference on Emerging Trends in Physical Sciences (NCETPS) Organized by Dept. Of Physics N.B.Mehta science college, BORDI 23rd and 24th Jan 2015.
47. Minal P.Patil and **T. N. Ghorude** “**Measurement of chromaticity coordinates of pH Indicator paper using RGB colorimeter and software techniques**” National Conference on Emerging Trends in Physical Sciences (NCETPS) Organized by Dept. of Physics N.B.Mehta science college, BORDI 23rd and 24th Jan 2015.

48. **T. N. Ghorude** and A. D. Shaligram “**Wireless Tristimulus Colorimetry for Skin Color Measurement**” National Conference on Emerging Trends in Physical Sciences (NCETPS) Organized by Dept. of Physics N. B. Mehta science college, BORDI 23rd and 24th Jan 2015
49. **T. N. Ghorude** and A. D. Shaligram “**Microcontroller Based Color Measurement Using RGB LED’s and TCS230 Colour Sensor**” Deccan Current Science Vol. 6 No. 1 January 2012 pp 395-398.
50. **T. N. Ghorude**, and A. D. Shaligram “**Microcontroller driven RGB LED system for tristimulus surface colorimetry**” 13th National seminar on Physics and Technology of Sensors (SPTS-13) organized by Department of Electronic science, Pune university, Pune, 3rd March to 05th March 2008.
51. **T. N. Ghorude**, Jinesh P. Shah and A. D. Shaligram “**Color Recognition Using Artificial Neural Network**” National Conference on Physics of Semiconductor Devices and Smart Materials Organized by Karmaveer Bhaurao Patil College Vashi, Navi Mumbai, 22-23 December, 2007
52. **T. N. Ghorude** and A. D. Shaligram “**Color recognition by using RGB LED cluster and software techniques**”, 12th National seminar on Physics and Technology of Sensors (SPTS-12) organized by **Bhabha Atomic Research Centre, Mumbai** 400085, 07th to 09th March 2007
53. **T. N. Ghorude** D. B. Gaikwad and A. D. Shaligram “**Microcontroller Based Color Measurement Using RGB LEDs**”, 11th National seminar on Physics and Technology of Sensors (SPTS-11) organized by Centre for Sensor studies and Dept. of Electronic Science University of PUNE- 411007, 27th Feb to 01st March 2006
54. **T. N. Ghorude**, D. B. Gaikwad and A. D. Shaligram “**Optical fiber Based color measurement by using RGB LED’s**”, National seminar on Interdisciplinary Applications of Electronics organized by Arts, science & commerce college Chopda, Dist. Jalgaon 28 – 29th, January 2006
55. **T. N. Ghorude** and A. D. Shaligram “**The CIE color matching functions & calculations of CIE chromaticity co-ordinates under C++**”, National level workshop on Device-grade Material Development Using Ion Beams, jointly Organized by Nuclear science centre, New Delhi and Department of Physics, University of Mumbai during 3rd and 4th November 2003.

11. BOOKS PUBLISHED:

1. “STUDIES OF MULTICOMPONENT AND MULTILAYER COATINGS” by **scholar’s Press Germany** ISBN-978-3-639-66223-8. Feb 2022
2. “Tristimulus wireless colorimeter and its medical Applications” by **scholar’s Press Germany** ISBN-978-613-8-91489-1. May 2020
3. “Tristimulus Colorimetry and its Applications” by **scholar’s Press Germany** ISBN-978-3-639-66613-7. Dec 2014
4. “Hand Book of Practical Physics for F.Y.B.Sc.” by **Sheth Publication** ISBN -978-81-942718-4-0. Apr 2016
5. “Hand Book of Practical Physics for F.Y.B.Sc. Revised Syllabus” by **Sheth Publication** ISBN -978-93-5149-421-8. Nov 2019
6. “Hand Book of Practical Physics for F.Y.B.Sc. Revised Syllabus” by **Sheth Publication** ISBN -978-93-5576-362-4. August 2022.
7. “Hand Book of Practical Physics for F.Y.B.Sc. Revised Syllabus” as per NEP 2020 by **Sheth Publication** ISBN -978-93-6268-213-0. August 2024.

12 MINOR RESEARCH PROJECTS :

- SIX minor research projects completed, sanctioned by **BCUD University of Mumbai**.
- One minor research project completed, sanctioned by **UGC-WRO Pune**.

List of the Projects completed.

Sr. No.	Title of the Project	Funding Agency	Sanction letter no.	Month and year	Amount
1.	Development of Tristimulus Colorimeter and its Industrial Applications	BCUD Mumbai University	Project No. 177	July 2008	Rs.30000/-
2.	Development of Microcontroller Based Colorimeter and its Medical Applications	BCUD Mumbai University	Project No. 96	June 2010	Rs.17000/-
3.	Development of Tristimulus Colorimeter and its Medical Applications	UGC	1375-F47-1379/10(WRO)	July 2010	Rs.1,10,000/-
4.	Wireless online Color matching solution using TCS 230 color sensor	BCUD Mumbai University	Project No. 15	July 2012	Rs.31400/-

5.	Development of wireless Tristimulus Colorimeter using TCS 230 color sensor for skin color measurement	BCUD Mumbai University	Project No. 160	July 2015	37000/-
6	Study of Corrosion Resistance Properties of CrN and Multilayer CrN/Cr Coating at Different Electrolyte Temperature Deposited on Stainless Steel by Vacuum Arc Process	BCUD Mumbai University	Project No. 381	July 2018	60000/-

13 MEMBERSHIP OF ACADEMIC ORGANISATION:

1. Life Member - **Indian Physics Association** (since 1997)
(Reg. No. – BOM/77/1971/BBSD Membership No. – BOM/LM/11500)
2. Life Member - Physics Teacher Association (Since 2000).
3. Life Member - **Gokhale Education Society**, NASIK-5
4. Member of **Board of Studies and Faculty of Science**, Member University of Mumbai from Dec 2010 to Dec 2015. Re-elected as BOS member with highest votes on 9th Feb 2018 to 2023 & Oct 2023 to Dec 2027.
5. Life member of International Science Community Association **ISCA-FM- 287** Dated 12 Sept 2013.
6. **CHAIRMAN** of BOARD OF STUDIES in PHYSICS University of Mumbai from Dec 2023 for five years.
7. **Member of Academic Council** University of Mumbai from Dec 2023 for five years.

14. ADDITIONAL MAJOR RESPONSIBILITIES :

- 1] From 01st July 2005 to 17th march 2006 worked as **In-charge Principal** and from 14th Aug 2006, working as **vice Principal (18Years)**.
- 2] **Working as, Head Department of PHYSICS** from 2005.
- 3] Local Managing Committee Member **CDC** (1995 till date).
- 4] Programme officer of **National Service Scheme** from 1995 to 2000 (Five Years)
- 5] **Management Representative for ISO** 9001 – 2008 & ISO 9001-2015 (24 Years)
- 6] University Exam: B.Sc. Ist to B. Sc. IIIrd year practical as well as Theory examiner and paper setter from last 25 Years.

7] I have been elected as **Chairman of Board of Studies (Physics), University of Mumbai** on 20th December 2023 and **Member of Academic council**, University of Mumbai. For three successive turns (15 years), Elected as **B.O.S. member**.

8] Working as **Reviewer of UGC-Care listed Journal** – **Mapana Journal of Sciences**

B.Sc., M. Sc. paper setter, examiner, Moderator and practical **Chairman in university of Mumbai**.

- Recognized as a **Post – Graduate teacher** of the University of Mumbai for the M.Sc. degree in the subject of PHYSICS by paper from 03rd November 2004 (Ref. No – PG/2/147 of 19th Jan 2005)
- Recognized as a **M.Phil. Guide** by YCMOU Nasik
- Recognized as a **Ph.D. Guide** by University of Mumbai w.e.f. 18th Jan 2013 Letter no. PG/2/ICD/2012-13/28991 on 6th Apr 2013
- **Four Students have been awarded Ph.D.** degree under my Guidance.
- **Three Ph.D. students are registered** under my Guidance, out of which **Two** has submitted Thesis.
- **MPSC** Prelim and Main Paper Setter and Examiner.
- Worked as subject expert nominated by VC for the interview of Assist. Prof. in various colleges.
- Worked as Joint chief conductor for University examination (letter no. COE/33/2013 dated 02nd Apr 2013) between 03rd Apr 2013 to 22 Apr 2013 at Sonopant Dandekar college, PALGHAR.
- Worked as a Judge for **“AVISHKAR”- a Research convention 2015-16** for P.G./PPG/Ph.D. level Project competition, on behalf of University of Mumbai on 21st Dec 2015 at C.K.T. College New Panvel.
- Worked as a **member of L.I.C** visited to **St. John College of Engineering & Technology** for continuation of Affiliation on 26th April 2016.
- An Article on the topic **“ANDROID”** is **published** in leading newspaper **DAINIK BHASKER** (National Paper) dated Friday, 20th Feb 2015.
- **Dr. Mrs. M. P. Patil & Dr Rajesh Mundotia successfully completed Ph.D.** on 26th December 2018 & 11 Dec 2021 under my guidance.
- Worked as **External Referee For Bangaluru University** and conducted Ph.D. Viva-Voce of Mr. Vasantharaju N. For The Topic “Study Of Dynamical Evolution of Solar Magnetic Fields” On 19th March 2021 At 2pm.

- Worked as member of Interview panel for admission of Ph.D. Govt. Of Maharashtra I.Y.College, Jogeshwari Mumbai on 28th Apr 2022
- Sathaye College Resource person – One day workshop for Orienting towards Physics Teachers for NEP 2020 Compliant syllabus and Curriculum on 28th Oct 2024 in collaboration with BOS & UGC-MMTTC

15. EVENTS ORGANIZED :

1. **Organizing Secretary** - **Two days** National Conference on “Emerging Trends in Physical Sciences (NCETPS)” Organized by Dept. Of Physics N. B. Mehta science college, BORDI 23rd and 24th Jan 2015.
2. **Organizing Secretary**- One day National seminar on “Role of Physics in daily life” Organized by Dept. Of Physics N. B. Mehta science college, BORDI
3. As **Vice –Principal**, organizing and attending various workshops and seminars and events like Blood Donation Camp, AIDS awareness programmes, Exhibitions and other social activities.

16. AWARDS:

1. International Institute of Educational and management and International Business council presented “**Bharat Gaurav Award**” for outstanding Achievements on 25th July 2016 at New Delhi.
2. **Lifetime Education Achievement award** was given to **Dr.T.N.Ghorude** by, **National & International compendium**, New Delhi on 8th March 2019.
3. **Best Teacher Maharashtra Rajya Shikshak Sena** Rajyastariya Gunvant Puraskar Sohala - Konkan Vibhag March 2020
4. National Level **Thakur Awards for Excellence** as “**Excellent Vice-Principal**” for the year 2021-22 instituted by TCSC in association with TRCAC, TRCL and TRPS. In the esteemed presence of the **Vice Chancellor Prof Suhas Pednekar** -University of Mumbai. Date: 27th Mar 2022 **Date:** 27th March, 2022 **Day:** Sunday **Time:** 5:00 pm
Venue: The Lalit, Andheri (E), Mumbai

17. PATENT:

- 1) I have got **Patent** for my research work regarding “**WIRELESS TRISTIMULUS COLORIMETER**” Publication Date: 05/05/2017 **Application No.201721013509 A**
- 2) I have got **UK Patent** on Title: “**Solar Powered Electric Vehicles Charging Station**” Publication Date: 26 February 2024 No. 6347534
- 3) Indian patent Filed on 12th Mar 2024 “**Development of microcontroller based Haemoglobin meter**”

18. PERSONAL DETAILS:

Father's Name : Ghorude Namdevrao Laximanrao
Permanent Address : A- 101, Prathmesh residency – I, near Parnaka, Dahanu 401601.
Date of Birth : 20th June, 1966.
Languages Known : **English, Marathi, Hindi, Kannada and Gujrathi**
Marital Status : Married
Cast : Hindu (Maratha)
Nationality : Indian
Mailing Address (Office) GOKHALE EDUCATION SOCIETY'S
N. B. Mehta Science College, Bordi,
Tal – Dahanu, Dist – Palghar, 401 701.
E-mail: tnghorude@gmail.com & nbmphysicshead@gmail.com
Mob. No. **9422684640 and 8080807273**

PROF. DR. GHORUDE TATYARAO NAMDEVRAO